DATABASE MANAGEMENT · PROJECT PROPOSAL

NYC 311 Explorer

A MySQL database and dashboard for finding patterns in city complaint data

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Problem & Goal

The problem

NYC 311 has logged about 23M service requests since 2020. The raw data is one huge 44 column excel table that is hard to search or compare.

Why it matters

Complaint data shows when and where problems happen and how fast the city responds. Today that is hard to see.

Expected results

A structured MySQL database, a dashboard to search and filter it, and patterns in time, place and city response.

Example questions: When do noise complaints peak? Which areas wait longest for a response?

What the Database does

Search

Find complaints by address, zip code or keyword.

Filter

Narrow by borough, complaint type, agency, date range, etc.

Reports

Counts by hour and month, response times, top problems and spike days.

User accounts

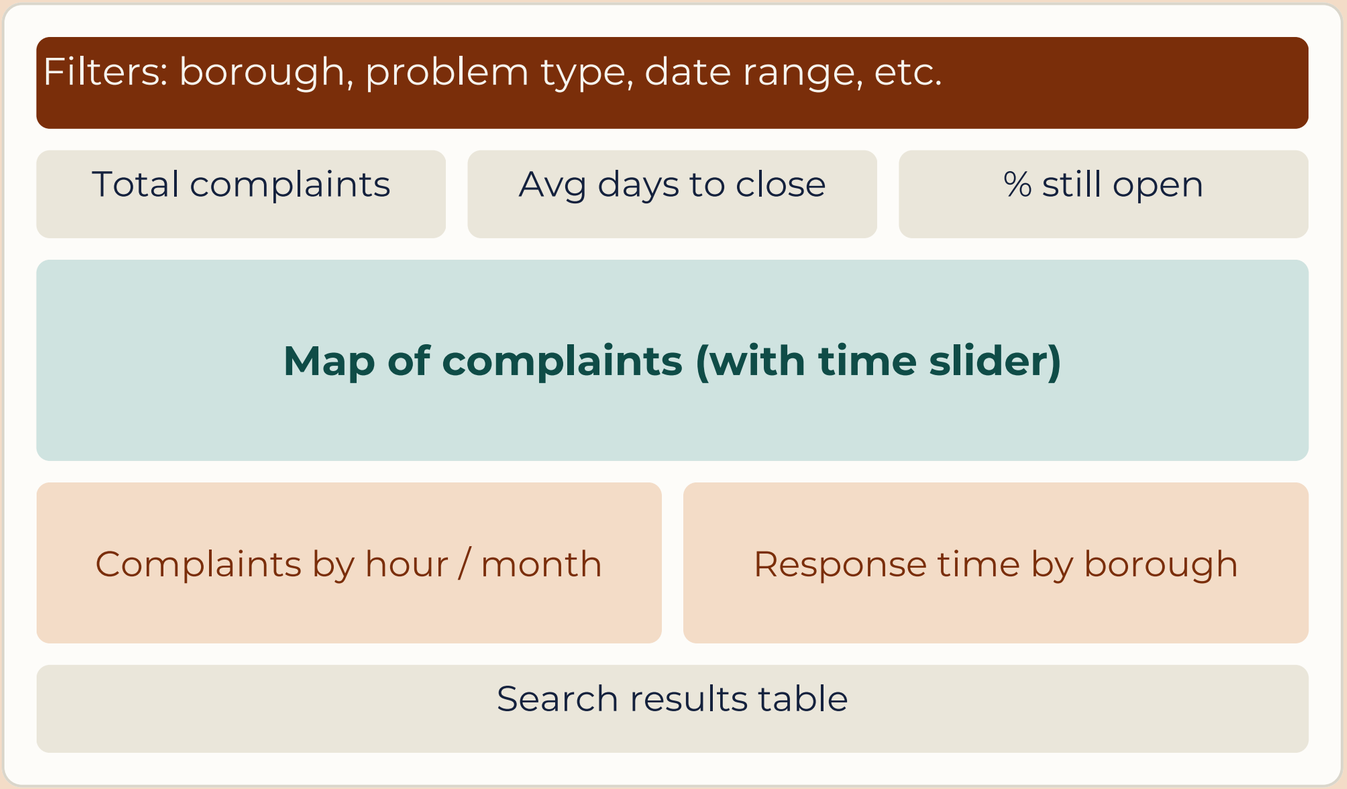
Sign in, save a filter combination and reopen it later.

Data and Database Design

Data	Details
Source	NYC Open Data: “311 Service Requests from 2020 to Present”
Scale	About 23M rows, 44 columns
Updates	Published daily
Loading	Download from NYC Open Data, load into MySQL, starting with a filtered subset

Table	What it holds
requests	Each complaint
problem_types	Problem and detail
agencies	Who responds
locations	Borough, zip, latitude, longitude
users	Dashboard accounts
saved_reports	Each user’s saved filters

Dashboard Plan



How we build it

- Python and Streamlit web app
- Each panel runs one SQL query on MySQL
- MySQL does the counting, so only small results reach the page
- Runs locally for the beta testing

Schedule

Week	Task	Owner
1	Choose complaint types to load, design tables and ER diagram	Data
2	Load the data into MySQL; add indexes	Data
3	Write and test SQL queries: time, place and response patterns	Queries
4	Build the dashboard: filters, charts, search table	Dashboard
5	Add user accounts and saved reports, map if time allows	Accounts
6	Test, compare query speed before and after indexes (if time allows)	All